Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/03/2014
PI Name:	Haswell, Elizabeth Ph.D.		
Project Title:	Mechanosensitive Ion Channels in Plants: Genetic, Computational and Systems-Level Approaches to Understanding their Proposed Role in Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	ehaswell@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314-935-9223
Organization Name:	Washington University in St. Louis		
PI Address 1:	Department of Biology		
PI Address 2:	1 Brookings Dr, Box 1137		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130-4862	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	07/31/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	June 2014 report: Kira Veley, PhD 50% effort ; Maggie Wilson, PhD 50% effort; Kelsey Kropp, MS 100% effort; Sarah Kloepper, undergraduate summer researcher supported by WUSTL SURF program; Samantha Embrick, undergraduate work-study		
COI Name (Institution):	Spalding, Edgar Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	NNX13AM55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	How plants respond to gravity is a fundamental biological question with important implications for space flight. Here we propose ground-based research designed to lead to space flight, in the area of plant biology. Our experimental approach focuses on several families of putative mechanoreceptors known as mechanosensitive (MS) ion channels. MS ion channels have been proposed to play a role in the perception of gravity in plants for many years, a hypothesis that is supported by several lines of evidence, but one that has not yet been systematically tested by molecular genetics. The research plan utilizes contemporary methodologies for the study of the model plant Arabidopsis thaliana to interrogate the biological function of plant MS ion channels and builds upon previous NASA-funded research into the response of plants to altered gravity environments. We will determine the role played by MS ion channels in mediating gravity signal transduction, their putative roles in adaptation to microgravity; and the molecular mechanisms by which the model MS ion channel MSL10 activates stress-response pathways in plant cells. We expect that these activities will have the following outcomes: 1) Test the long-standing hypothesis that MS ion channels function in gravity perception; 2) Contribute to our basic knowledge regarding how plants exploit MS ion channels to sense and respond to mechanical stress; 3) Provide information that could be used on Earth to improve crop productivity; and 4) Establish a foundation of knowledge required to generate more mechanistic and sophisticated hypotheses about plant gravity perception that can be tested in spaceflight using the European Modular Cultivation System and/or the ABRS GFP Imaging system. (These proposed studies directly address the objectives of the solicitation, as they aim to answer basic questions about how plants respond to changes in gravity and should lead to hypotheses that can be tested on future space flights.)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Identifying the molecular mechanisms by which MS ion channels function and evaluating their roles in the perception and response to altered gravity conditions will help fill a large gap in our knowledge regarding the molecules involved in gravity and mechanoperception. Gaining this knowledge has the potential to achieve two long-term goals: 1) to mitigate the effects of mechanical stress on crop plants on Earth, and 2) to counteract the effects of altered gravity environments on plants grown in space. These experiments will address the high priority goal of “understanding the basic mechanisms of plant response” to changes in gravity described in the Decadal survey of 2011.
Task Progress:	Research Progress: Aim 1: Test the Hypothesis that MS Ion Channels Play a Role in Root Gravitropism Using a Machine Vision and Computation Approach Mechanosensitive (MS) ion channels have been implicated in a variety of steps in gravity signaling, but are most often implicated in the initial gravity response. MS ion channels have been proposed to sense ER membrane tension as a result of the amyloplasts sedimentation; the plasma membrane tension resulting from the uneven pressure of the tonoplast at the bottom of the cell; or the action of amyloplasts on actin microfilaments tethered to the plasma membrane. There are at least three potential barriers to the identification of MS ion channels in forward genetic screens for genes involved in gravity response: 1) redundancy of MS ion channel gene function; 2) redundancy of gravity response pathways involving MS ion channels; and 3) subtle effects of MS ion channel function on the gravity response. We aim to test the long-held hypothesis that MS ion channels mediate the perception of changes in gravity vector by using a reverse genetics approach and a highly sensitive Machine Vision and Computation assay for defects in gravity perception. Progress: We have collected a number of mutants in the model plant Arabidopsis thaliana implicated in gravitropism from other labs and stock centers and collected several mutants harboring lesions in known or predicted MS channels. We have started crossing these lines to make higher order mutants in preparation for gravity testing. Once established these lines will be sent to Edgar Spalding at U. Wisconsin (a co-PI on this grant) for root gravitropic analysis and we will analyze shoot gravitropism and other phenotypes here at WUSTL. One line we have already isolated, which lacks MS channels in the plastid envelope, shows defective root and shoot gravity response. We are following up on this observation to determine if this is due to defects in amyloplast sedimentation or in a signaling pathway. We have made progress in instrumentation. Postdoc Maggie Wilson machined and assembled (with the help of her father) a set of polycarbonate plate trays that will help us keep our agar plates growing just a few degrees from the vertical, and we have purchased a microscope from and are right now working on setting it on its back to serve as a vertical stage microscope. We are exploring the possibility of producing a support for the microscope with a 3-D printer. Alternatively, the WUSTL machine shop has planned a support made of the usual metal components. Aim 2: Test the Hypothesis that MSL and Other MS Ion Channels Are Involved in the Perception and/or Adaptation to Weightlessness. Progress: We do not have progress to report on this aim. Aim 3: Determine the Molecular Mechanism of MSL10-Induced Reactive Oxygen Species (ROS) Signaling MS ion channels are implicated in the perception and/or response to changes in gravity conditions. The experiments in this aim are designed to gain a more molecular and mechanistic handle on the regulation and signaling behavior of MS ion channels, building on our recent observations that overexpressing the MscS-Like MS ion channel MSL10 from A. thaliana leads to ROS-associated cell death. Understanding how MSL10 is regulated and how its activation leads to ROS-related stress signaling may be important for mitigating the effects of mechanical stress on crop plants or plants in the space environment. We have found that MSL10 induces cell death signaling in a phosphorylation-dependent manner and that the N-terminus of MSL10 alone can provide this activity, independent of the ion channel function of the channel. We continue to suspect that 14-3-3 phospho-binding proteins (also called GRFs) may play a role in regulating MSL10 activity and are working on testing this idea. We have been trying to develop an assay for ROS production in response to osmotic shock. Our initial experiments growing seedlings in liquid MS + 50 mM NaCl then measuring cell death and H2O2 production have not been successful, but we have a number of ideas about ways to improve our experiments, and will be actively pursuing this approach. Once developed, such an assay could help us identify the kinase and phosphatase involved, and develop a time course of gene expression response to osmotic shock and ROS production.

	We have also initiated a series of experiments to address the role of ROS production in msl2 msl3 mutants (these are the MS channels in the chloroplast membrane). We are pursuing a number of lines of evidence that suggest that a signaling pathway associated with the phytohormone cytokinin may induces ROS production and callus formation in this mutant plant. Experiments are ongoing to test this hypothesis. Two journal articles have been submitted for publication.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Abstracts for Journals and Proceedings	Haswell E. "Up and Away: Statolith Morphology Alters Gravity Response in the Model Plant Arabidopsis thaliana." Presented at 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 77. , Nov-2013